Cristoff Stan

Senior LLM Engineer - RAG, Agents and Evaluation

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PROFESSIONAL SUMMARY

Senior LLM Engineer with 10+ years delivering B2B SaaS, e-commerce, and enterprise operations platforms using Python, LLM APIs, open-source models, RAG, vector databases, agent workflows, evaluation; combines production AI architecture, legacy modernization, cloud-native delivery, performance engineering, and technical leadership to build secure, scalable systems that improve reliability, release velocity, operational efficiency, and measurable business outcomes.

TECHNICAL SKILLS

- **LLM Engineering:** Python, OpenAI-compatible APIs, Anthropic APIs, Hugging Face Transformers, open-source LLMs, structured outputs, function/tool calling and prompt/version management.
- **RAG and Search:** Embeddings, chunking, hybrid retrieval, reranking, pgvector, Pinecone, Weaviate, Milvus, Elasticsearch/OpenSearch, knowledge graphs and citation grounding.
- **Agents and Orchestration:** LangChain, LlamaIndex, LangGraph, tool execution, memory patterns, workflow state, retries, idempotency, human approvals and multi-agent coordination.
- **Backend and Integration:** FastAPI, Django, Node.js, REST, GraphQL, WebSockets, streaming responses, background workers, event-driven architecture and document processing.
- **Evaluation and Safety:** Offline evaluation, calibrated LLM-as-judge, retrieval metrics, hallucination checks, red teaming, guardrails, PII handling, content safety and audit logs.
- **Cloud and MLOps:** AWS Bedrock/SageMaker/EKS, GCP Vertex AI/GKE, Azure OpenAI/AKS, Docker, Kubernetes, Terraform, model gateways and cost controls.
- **Data and Messaging:** PostgreSQL, pgvector, Redis, Kafka, object storage, ingestion pipelines, metadata and access-control filtering.
- **Senior Engineering Practices:** Architecture reviews, technical leadership, mentoring, code reviews, legacy modernization, performance engineering, production readiness, secure development, Agile delivery, technical estimation, stakeholder communication and cross-team release coordination.

PROFESSIONAL EXPERIENCE

BEVY — **September 2021 - June 2026**

Senior Software Engineer — Remote, Romania

- Architected **multi-tenant B2B SaaS and enterprise operations platforms** using **Python 3.8-3.12, PyTorch 1-2, TensorFlow 2, FastAPI, vector databases, Kubernetes, and cloud AI services**, defining scalable domain boundaries, reusable delivery standards, and production controls across several product teams.
- Modernized tightly coupled legacy applications into **modular, cloud-ready services and interfaces** aligned with **Python, LLM APIs, open-source models, RAG, vector databases, agent workflows, evaluation, guardrails and cloud deployment**, using staged migration, compatibility APIs, feature flags, and contract validation to protect active customers.
- Reduced **production inference latency by 34%** through model/runtime profiling, batching, caching, asynchronous processing, optimized retrieval, and autoscaling policies.
- Resolved production incidents involving **database-pool exhaustion, memory growth, duplicate background jobs, stale cache entries, partial downstream failures, and unbounded retries** by adding lifecycle controls, idempotency, correlation IDs, and failure isolation.
- Led safe upgrades across **Python 3.8-3.12, PyTorch 1-2, TensorFlow 2, FastAPI, vector databases, Kubernetes, and cloud AI services** by replacing unsupported dependencies, updating build images and deployment manifests, validating breaking changes through regression and load testing, and releasing through canary stages with rollback plans.
- Delivered new capabilities for **billing, subscriptions, reporting, audit history, notifications, search, administration, and customer-support workflows**, translating complex requirements into secure APIs, resilient asynchronous processing, and maintainable user experiences.
- Improved architecture with **Domain-Driven Design, clean architecture, event-driven workflows, transactional outbox processing, tenant-aware authorization, distributed caching, and observability**, strengthening consistency across synchronous and asynchronous operations.
- Strengthened **CI/CD and production quality** with **Docker, Kubernetes, Terraform, automated tests, security scanning, schema-migration gates, deployment approvals, and AWS, Azure, and GCP monitoring**, reducing release risk and improving operational visibility.
- Mentored engineers through **architecture reviews, pull-request feedback, production debugging, incident retrospectives, estimation, and release-readiness sessions**, improving code consistency, technical ownership, and cross-team delivery decisions.

Syscom Digital
October 2019 - August 2021

Software Engineer — Remote, Romania

- Developed commerce, customer-service, and operational products using **Python 3.6-3.9, TensorFlow 1-2, scikit-learn, Flask, SQL, and cloud data services**, supporting checkout, order tracking, refunds, fulfillment, reporting, user administration, and third-party provider integrations.
- Implemented new **payment, notification, search, workflow, export, and reconciliation features** with secure APIs, asynchronous workers, validation, and audit-friendly state transitions across customer and administrative experiences.
- Reduced **model and integration defects by 28%** through reproducible datasets, versioned experiments, service-contract tests, evaluation baselines, and staged model promotion.
- Investigated production issues involving **duplicate provider callbacks, inventory mismatches, stale sessions, timeout chains, malformed payloads, and long-running database transactions**, then introduced concurrency controls, bounded retries, and reconciliation routines.
- Modernized older modules within **Python 3.6-3.9, TensorFlow 1-2, scikit-learn, Flask, SQL, and cloud data services** by introducing typed contracts, clearer service boundaries, shared components, supported runtime versions, and compatibility tests that enabled gradual replacement without interrupting daily operations.
- Improved performance through **server-side pagination, indexed queries, reduced serialization, selective data loading, caching, connection-pool tuning, and browser or runtime profiling**, keeping high-volume administrative workflows responsive.
- Expanded delivery pipelines with **Jenkins/GitLab CI, Docker, unit and integration tests, static analysis, artifact versioning, migration validation, smoke tests, and staged environment promotion** across development, QA, and production.
- Collaborated with product owners, designers, QA, support teams, security reviewers, and external providers to convert ambiguous requirements into **testable stories, resilient failure handling, measurable acceptance criteria, and release documentation**.

Soft Build
June 2016 - July 2019

Software Developer — Remote, Romania

- Built full-stack business applications using **Python 3.4-3.8, scikit-learn, OpenCV, pandas, NumPy, and batch analytics** for commerce, booking, pricing, inventory, reporting, account management, and internal administration across multiple client engagements.
- Reduced **training and batch-scoring time by 32%** by vectorizing preprocessing, caching reusable features, improving data loading, and eliminating redundant model execution.
- Developed new **REST endpoints, reusable interface modules, scheduled imports, reporting features, role-based workflows, and provider integrations**, keeping validation and business rules separated from delivery and persistence concerns.
- Resolved defects involving **duplicate submissions, stale inventory, null-reference failures, inconsistent validation, slow database queries, missing error feedback, and partial updates** through transactions, defensive checks, indexed access, and explicit recovery states.
- Moved blocking email, document, synchronization, import, and report-generation work into **background jobs and queue-based processing**, adding retry policies, schedules, failed-job diagnostics, and operational visibility for support teams.
- Expanded quality coverage with **unit, integration, API regression, browser, and database migration tests**, prioritizing authentication, pricing, persistence, concurrency, and other high-risk customer workflows.
- Supported **execution-plan analysis, production incident resolution, deployment automation, code reviews, technical estimation, and legacy modernization**, while introducing reusable architecture and coding standards across projects.

Decima Digital
November 2015 - April 2016

Software Developer — Remote, Estonia

- Delivered customer-facing websites and administration tools using **Python scripts, SQL, text/image processing utilities, and rule-based automation**, supporting account workflows, content management, file handling, search, forms, notifications, and operational reporting.
- Maintained server-rendered and tightly coupled legacy modules while introducing **reusable services, clearer controllers, modular client code, and consistent validation**, allowing new features to coexist safely with older functionality.
- Built controllers, service classes, repositories, scheduled utilities, and database scripts for **content retrieval, form processing, account operations, file uploads, notification routing, and report generation**.
- Fixed production defects involving **failed uploads, broken validation, malformed requests, null-reference exceptions, duplicate submissions, database connectivity errors, and missing user feedback** through reproducible diagnostics and defensive handling.
- Improved slow operations by reviewing generated SQL, removing unnecessary object loading, adding targeted indexes, reducing repeated in-memory loops, and replacing oversized queries with focused data access patterns.
- Participated in **deployment preparation, browser and API compatibility testing, backup verification, smoke testing, rollback planning, and requirement reviews** with designers, project managers, and senior engineers.

Borna Technology | **September 2014 - August 2015**

Software Developer Intern | Tallinn, Estonia

- Assisted senior engineers with **Python 2.7, data preparation scripts, SQL, and basic statistical analysis** across internal tools and client-facing applications, following established coding, source-control, database, and release practices.
- Built small interface enhancements, reusable templates, server-side handlers, validation scripts, and database-backed administration pages while learning structured debugging and maintainable implementation patterns.
- Created and updated **SQL queries, stored procedures, migration scripts, test data, and integration utilities** under supervision, applying safe database-change and review practices.
- Investigated **invalid submissions, null-reference errors, database connectivity failures, layout defects, inconsistent API responses, and deployment configuration problems** by reproducing issues and documenting diagnostic steps.
- Contributed to **QA checks, deployment notes, smoke tests, code reviews, and technical documentation**, establishing a practical foundation in collaborative delivery, production support, and release discipline.

SELECTED PROJECTS

Production AI Service Platform

Designed a secure, observable AI platform using Python, LLM APIs, open-source models, RAG, vector databases, agent workflows, evaluation, guardrails and cloud deployment, supporting data ingestion, evaluation, model or retrieval workflows and real-time application integration.

Implemented versioned artifacts, automated evaluation, access controls, guardrails, workload scaling, traceable outputs and rollback-safe deployments.

AI Workflow Modernization Program

Led the transformation of experimental notebooks and scripts into tested services, reproducible pipelines and monitored production workflows.

Resolved inconsistent datasets, fragile prompts/models, limited evaluation and opaque failures through versioning, regression suites, telemetry and human review controls.

EDUCATION

Tallinn University - Bachelor's Degree in Computer Science | September 2011 - May 2014